

AI-Generated Steganographic Malware: Evaluating Detection and Evasion in Modern Anti-Virus Systems

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1. CHAPTER 1

1.1. Introduction

Alotaibi et al. (2024) state that we live in an era increasingly defined by digital imagery and advanced artificial intelligence. With large language models (LLMs) improvements a subtle yet potent cyber threat is quietly escalating, AI-generated steganographic malware (Gao et al., 2022). Steganography has traditionally been a practice of hiding messages and use of invisible inks in art to conceal information (Chang & Echizen, 2025). New steganographic methods have emerged merging the sophisticated discipline into cybersecurity (Chang & Echizen, 2025). This form of attack leverages the very technologies designed to enhance our digital lives to conceal malicious code within seemingly innocuous files, rendering traditional cybersecurity defences often powerless (Motwani et al., 2024). This advancement not only compromises the authenticity and reliability of digital visual media but also erodes user trust, as seemingly harmless images can become conduits for sophisticated cyberattacks (Deng et al., 2024). With modern security breaches perimeter defences and insider threats are increasingly bypassed and causing more disruption in business operations (Rajendran et al., 2019).

This study aims to improve the detection capabilities of open-source antivirus solutions by focusing on YARA, a widely used rule-based malware detection tool (NCCIC, 2015). Antivirus software are most used tools in protecting users' computers from malware (Pérez-Sánchez & Palacios, 2022). An investigation by Almomani et al. (2022) found that steganographic malware is particularly challenging for modern antivirus software to analyse. In their study, they tested 65 antivirus engines with a ransomware sample. While 35 of the engines were able to detect the ransomware before it was hidden, none could identify it after it was concealed within a video frame. This result illustrates the significant detection challenges that steganographic malware poses to current security tools. This study builds on those findings without evaluating specific antivirus software. An experiment done by Mirza et al. (2021) has proven that, where a malicious PNG file was embedded within a benign application, it evaded all antivirus detection on Virus Total which means that steganographic techniques greatly improve malware evasion.

This study will evaluate the limitations and capacity of YARA for detecting AI-generated steganographic malware. It will build on the work of Hammadeh and Kavitha (2023) by using machine learning and custom YARA rules to develop a strategy that improves identification and reduces false positives (Bhardwaj & Dave, 2023) This study aims to improve the detection of AI-generated steganographic threats in open-source antivirus software using YARA rules and ML anomaly

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detection. The resulting methods will be robust and designed for easy integration into existing tools. The findings will benefit the cybersecurity community and further strengthen steganographic malware detection mechanisms, ensuring that AV systems remain effective against evolving cyber threats.

1.2. Concepts Central to the Study

This section introduces the fundamental concepts that explains this study. This research contains numerous important concepts that will underpin our understanding, and its progression as outlined (pressbooks, 2015).

As the study advances, the concepts just do not become definitions, but they become an integral component of the broader study, serving as the foundation for understanding the study (Pearl, 2009). The notions will help the study understand the study's progress and avoid pitfalls during the investigation:

1.2.1. Steganography

Steganography is the art of concealing hidden information making it difficult to detect (Rahman et al., 2022). The rise in the use of steganography has been influenced by the improvements in information digitalization and data transfer (Majeed et al., 2021). Although the use of steganography can be for good cause such as hiding of confidential information online that can be exploited by criminals (Rahman et al., 2022), it has also grown into a field where parties responsible for cybercrime abuse it to hide, transmit messages and also malwares (Rustad et al., 2023).

This study looks at the steganography application in concealing Morden AI generated malwares. Steganography has manifested in various forms, including visual, auditory and linguistic content (Chang & Echizen, 2025). The Linguistic technique is used to hide the message within the cover text in non-obvious way such that the presence of message is imperceptible to an outsider linguistic content (Kaur & Behal, 2014). Kaur and Behal (2014) further specify how the visual and auditory can be regarded as technical steganography where message is hidden in image, video, audio, text, or some other digital media.

The study will explore the technical steganography technique specifically the image digital media which makes use of the least significant bit (LSB) which allow the replacement of information from the lower image bits that hold no significant information about an image and replace them with the generated malwares (ALabaichi et al., 2020). Exploring both the insertion and substitution methods of the LSB, The Insertion based will allow the embedment of data increasing the image size and the substitution based method will allow the replacement of the bits

of the image with secret data without increasing the size of the image (Aslam et al., 2022).

ALabaichi et al. (2020) shown on their study that the host image and Steg malware embedded images do not present significant differences in visual when using the LSB method and proposed the algorithm used as it can successfully hide secret messages in host images without any distortion. However, Gupta et al. (2012) showed how as more bits are used from an image, the quality of the image deteriorates but this does not affect the quality of the message been hidden in the images.

1.2.2. Artificial intelligence

According to Jiang et al. (2022), AI as enabling machines to do things that require human intelligence. AI techniques such as genetic algorithms, particle swarm optimization, and artificial bee colony algorithms can be used to intelligently select carrier images used to conceal secret data and precise embedding locations within those images (Shyla et al., 2021). These approaches improve stealth and make detection by attackers much more difficult (Georges & Magdi, 2020).

A subset of Artificial Intelligence, Machine learning, will be leveraged in this study to be able to study the produced outputs and determine any anomalies which will assist in creating more targeted rules.

1.2.3. Machine learning-based Anomaly detection

Machine learning as defined by Arthur Samuel who created the checkers program, is the field of study that gives computers the ability to learn without being explicitly programmed (Mahesh, 2020). It is one of the most growing fields in today technology world and as Jordan and Mitchell (2015) further specified, Machine learning addresses the question of how to build computers that improve automatically through experience and it involves learning of hidden patterns within the data and subsequently using the patterns to classify or predict an event related to the problem (Alloghani et al., 2020).

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Machine learning algorithms can either be supervised or unsupervised (Alloghani et al., 2020) and this study will make use of the unsupervised machine learning. This subset of ML will allow us to develop models that detect trends in historical data for our steganography malware embedded images and utilize those models to make forecasts to generate data-driven prediction on the malware detection (Naeem et al., 2023).

Using unsupervised learning will be beneficial in finding patterns and similarly classify data points for our malwares specifically using anomaly detection to find outliers in the data points (Naeem et al., 2023) between the steganography

embedded images and their original images thus integrating them with YARA rules to enhance detection. Stoian (2020) on his study showed how Anomaly detection is great for spotting malware because it can quickly identify unusual behaviour in a system and through the use of different algorithms proved that it is a reliable way to catch threats that might otherwise go unnoticed such as steganography (Gao et al., 2022), making it a strong choice for keeping networks safe.

1.2.4. YARA

YARA is a tool widely used in cybersecurity to identify and classify malware which rely on the creation of custom rules that match specific patterns within files or memory which can be used for threat hunting (Vinay Patil, 2025). YARA is valuable because it helps detect malicious files effectively, especially in network environments where sharing threat intelligence can be challenging (Gazo et al., 2024). By using powerful open-source tools like Zeek and YARA, organizations can identify malware without needing expensive solutions making YARA a cost-effective and practical choice for improving cybersecurity (Coscia et al., 2025).

This study will leverage YARA signature-based rule creation to create rules that specifically target steganography malware in images. Signature-based detection is used by antivirus and malware analysis software to detect malware by comparing a repository of file signatures for known malware with signatures of files stored on a computer (Lockett, 2021). This approach is crucial because YARA's customizability allows for the development of rules that match specific patterns within files or memory, which can be invaluable for threat hunting (Khalid et al., 2020).

The ability to craft rules tailored to the unique characteristics of steganographic malware, such as unusual pixel changes, shifts in entropy, or changes in file structure, is essential for its detection (Khalid et al., 2020). While YARA is a powerful tool for detecting known malware, it is important to acknowledge its limitations, particularly against polymorphic malware that changes its patterns to evade detection, as it is a signature-based technique (Altawil & Youssef, 2023). Therefore, this study will combine YARA rules with machine learning-based anomaly detection to overcome these limitations and enhance detection capabilities for both known and unknown steganography malware.

1.3. Problem Statement

Steganography has traditionally been a practice of hiding messages and use of invisible inks in art to conceal information (Chang & Echizen, 2025). As AI continues to evolve, especially with large language models (LLMs), new steganographic methods have emerged this sophisticated discipline into cyber security challenge (Chang & Echizen, 2025). These methods allow the hiding of

AI-generated malicious code in shared images. The increasing use of AI generated steganography presents a significant cybersecurity challenge, as traditional antivirus software struggles to detect malware embedded within multimedia files (Gao et al., 2022). Antivirus software are most used tools in protecting users' computers from malware (Gao et al., 2022) but these technologies can sometime fail to deliver due to complexity or the advanced nature of the malware they combat (Jardine, 2020). Studies have shown that while some antivirus engines detect malware before it is embedded, they fail once it is concealed within an image or video frame as a stegomalware (Almomani et al., 2022).

This study applies machine learning-based anomaly detection with custom YARA rules and hypothesizes that it will yield better results in detecting AI-generated steganographic malware while reducing false positives (Hammadeh & Kavitha, 2023). YARA is a rules-based malware detection tool for open-source antivirus software (NCCIC, 2015). Systems that use YARA for rules detection suffer from the same weaknesses as other signature-based techniques as they cannot identify malware that changes their patterns to escape detection (Mirza et al., 2021).

The need for threat detection extends beyond known steganographic malware, which tools like YARA can identify, to encompass unknown threats as well, this study will propose a hybrid model with YARA to identify known steganography malware, and an ML-based anomaly detection model to detect unknown malwares. The study will follow two steps: First step being YARA-based signature detection, which scans files for predefined patterns (Rajendran et al., 2019) integrated with certain criteria tracing files identified as steganographic malware and the second being machine learning-based anomaly detection system, which analyses flagged files for statistical anomalies (Rivera et al., 2021), like unusual pixel changes, shifts in entropy, or changes in the files' structure (R.Loganathan, 2024). The target of this proposed process is to maximize detection of AI-generated steganographic malware and minimize other false alarms.

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1.4. Research Aims, Questions and Objectives

1.4.1. Primary Aim

In this study, the research aims and objectives play connected roles in shaping the investigation into the detection of AI-generated steganographic malware (Creswell & Creswell, 2017). The primary aim of this study is to evaluate and enhance the detection capabilities of open-source antivirus solutions against AI-generated steganographic malware, with a focus on transparency, customization, and community contribution to security improvements.

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1.4.2. Research Questions

Research questions are fundamentally based on answering important questions without bias and with precision (Mayo et al., 2013). Below are the research questions for this study:

- What are the current limitations of existing open-source antivirus solutions in detecting various types of AI-generated steganographic malware?
- To what extent can current YARA rules effectively identify AI-generated steganographic malware, and What are the common patterns of detection success and failure?
- How can customized YARA rules be developed to specifically target steganographic indicators for AI-generated malware?
- How can machine learning-based anomaly detection models be developed and implemented to identify statistical irregularities of AI-generated steganographic malware in files?
- How effectively can a hybrid detection system, integrating customized YARA rules and machine learning-based anomaly detection, improve the detection rates of AI-generated steganographic malware and reduce false positives compared to standalone methods?
- What performance benchmarks can be established for the developed hybrid detection system against AI-generated steganographic malware samples, and how do these compare to existing detection solutions?

These detailed research questions will guide this study in thoroughly investigating the evolving threat of AI-generated steganographic malware. This research aims to develop and evaluate robust detection methodologies, and the findings will ultimately contribute to enhancing cybersecurity defences against these threats.

1.4.3. Research Objectives

1.4.3.1. Investigate existing AV limitations

To investigate the limitations of the existing AV solutions in detecting steganographic malware. This will include testing against various types of AI-generated steganographic techniques to establish a baseline for detection capabilities.

1.4.3.2. Analyse YARA efficiency

To analyse the efficiency of YARA in identifying AI-generated steganographic malware. This involves testing existing YARA rules against known samples and identifying patterns of detection success and failure.

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1.4.3.3. Develop custom YARA rules

This involves creating new YARA rules that are designed to look for unique patterns or characteristics found in steganographic malware, making the detection more precise. While these rules will be used empirically, the act of developing them is a theoretical, conceptual exercise based on the steganographic indicators.

1.4.3.4. Build ML-based anomaly detectors

This means building machine learning models that can spot unusual or abnormal patterns in files that might indicate the presence of hidden malware, even if the specific signature isn't known.

1.4.3.5. Integrate into a hybrid pipeline

This final step combines the strengths of both the customized YARA rules (for known threats) and the machine learning anomaly detection (for unknown threats) into a single, more robust system.

1.4.3.6. Benchmark hybrid performance

Test the detection system against steganographic malware samples to validate improvements and establish performance benchmarks compared to existing solutions. It begins by tracing the evolution of steganography from its historical roots to its modern, AI-driven form (Nair & Rohini, 2025). The review then critically assesses the limitations of current antivirus solutions and the specific detection challenges posed by steganographic malware. It also evaluates the capabilities and shortcomings of YARA as a detection tool and concludes by exploring emerging solutions and research, setting the stage for the proposed hybrid detection model.

1.5. Literature Review

This literature review provides the foundational context for this study (Schryen et al., 2015) by examining the current body of knowledge on steganography and malware detection.

1.5.1. Evolution of Steganography

Hammad et al. (2024) shows how steganography has evolved from simple methods of hiding messages in physical media to using techniques like Least Significant Bit (LSB) substitution, which allows secret data to be concealed within image pixels without perceptible changes. All these enhance the capacity and robustness of data hiding, making digital steganography a powerful tool for secure and covert communication as mentioned by (Şahin et al., 2021).

The transformation of steganography from historical practice to modern cyber threat represents a significant shift in the security landscape (Bamanga & Babando, 2024). Chang and Echizen (2025) further document this evolution, highlighting how steganography has progressed beyond simple hiding to incorporate sophisticated AI agents that can generate and embed malicious code in ways that appear natural and unsuspecting. Their research emphasizes how traditional detection methods struggle against these advanced techniques.

Further expanding on this evolution, Gao et al. (2022) explore how AI-generated photorealistic images provide new vectors for steganographic secret sharing. Their work demonstrates how generative adversarial networks can be used to create images that contain hidden information while maintaining visual fidelity that passes human inspection, creating significant challenges for automated detection systems.

1.5.2. Limitations of Current Antivirus Solutions

The effectiveness of current antivirus solutions against sophisticated threats and new-generation attacks and evasion techniques, remains a critical area of concern (Aslan et al., 2023). Pérez-Sánchez and Palacios (2022) conducted comparative evaluations between local security event management systems and standard antivirus software, identifying significant gaps in detection capabilities against advanced concealment techniques. Their research highlights how conventional signature-based detection often fails against modified threats.

Jardine (2020) provides a more theoretical framework for understanding these limitations, exploring concepts of risk homeostasis and information problems in cybersecurity. This work explains how antivirus solutions can sometimes fail despite technological advances due to fundamental challenges in threat identification and evolution of attack methodologies. This research suggests that purely technical solutions may be insufficient without corresponding adaptations in detection.

1.5.3. Detection Challenges for Steganography Malware

The specific challenges of detecting steganographic malware are well-documented in recent literature. With Alazab and Pham (2023) finding that existing anti-malware, the signature-based tools and end point security solutions tend to be ineffective when analysing the concealed malicious content in multimedia files, and thorough byte level analysis. Almomani et al. (2022) further demonstrated a novel ransomware hiding model using HEVC steganography that successfully evaded detection by all tested antivirus solutions. Their work showed that even when

ransomware was detected in its original form, concealment through steganographic techniques rendered it invisible to conventional scanning methods.

Building on these findings, Hammadeh and Kavitha (2023) advocated for advanced machine learning algorithms as a necessary evolution in malware detection. Their research emphasized the need to move beyond signature-based approaches toward more adaptive systems capable of identifying anomalous patterns that might indicate concealed malicious code.

1.5.4. Yara Capabilities and Limitations

YARA represents a significant tool in the malware detection that uses patterns based on static and dynamic information (Ningappa, 2020). The National Cybersecurity and Communications Integration Centre provides foundational documentation on YARA's application for malware detection, highlighting its flexibility for custom rule development but acknowledging limitations against evolving threats (NCCIC, 2015).

The effectiveness of Yara against steganographic threats remains inconsistent as Mirza et al. (2021) explored these limitations by documenting specific malware evasion techniques that successfully bypass anti-malware solutions employing YARA rules. Their research demonstrates how traditional signature-based approaches struggle against concealment techniques, creating a need for more sophisticated detection methodologies.

1.5.5. Emerging Solutions and Approaches

Recent research has begun exploring potential solutions to these detection challenges. Rivera et al. (2021) investigate machine learning-based anomaly detection in real-time data streams, demonstrating potential applications for identifying concealed malicious code based on statistical irregularities rather than known signatures. This approach shows the possibility of creating a technique for detecting previously unknown threats.

R.Loganathan (2024) focused on anomaly-based malware detection using machine learning techniques, providing frameworks for identifying malicious code through behavioural and structural anomalies analysis suggesting that machine learning approaches can enhance traditional signature-based detection to create more reliable detection technique.

The security landscape continues to evolve as new emerging threats comes in and Rajendran et al. (2019) documented on how modern security threats like steganography can be deployed in a IoT environments through connected devices.

Their research emphasises on the fast-expanding attack surface created by the current connected devices.

Groenveld (2024) further provides a practical guidance on YARA rule and signature-based threat hunting strategies, suggesting methodologies for improving detection capabilities through more improved rule development. This work offers a foundation for enhancing YARA implementations specifically targeting steganographic malwares.

1.6. Research Methodology

This study will use an experimental design and analysis to evaluate detection methodologies against AI-generated steganographic malware. The research will follow a primarily quantitative approach where an objective reality independent of any observation (Almalki, 2016), measuring detection rates, false positive ratios, and processing efficiency across various detection techniques. This will be accompanied by qualitative assessment which places emphasis upon exploring and understanding (Almalki, 2016) of detection mechanisms and evasion techniques to provide contextual understanding of performance metrics.

The methodology incorporates a mixed-methods approach that includes:

- Technical analysis of steganographic malware samples
- Performance evaluation of detection systems
- Comparative assessment against benchmark datasets
- Iterative development and testing of enhanced detection methodologies

While the primary focus will be on quantitative metrics like detection rates and false positives to measure the technical performance of the developed system, qualitative assessment provides crucial context.

Qualitative methods will help in clearing why certain evasion techniques are effective and how specific detection mechanisms are performed under different conditions. For instance, observing the behaviour of the system against polymorphic steganographic malware (qualitative) can inform the development of more adaptive YARA rules or machine learning features (quantitative measurements).

1.6.1. Research Approach

The research will progress through four phases, each building upon findings from previous stages to develop a comprehensive detection solution. This phased approach ensures a structured and iterative development process, allowing for the refinement of techniques and integration of insights gained from earlier stages into subsequent ones (Wynn & Eckert, 2017). It starts with an initial analysis of existing tools to identify shortcomings, then moves to signature development, followed by

the integration of machine learning for anomaly detection, and finally, the implementation of a hybrid system that combines these strengths.

Phase 1: Analysis of Existing Solutions

The phase will evaluate the traditional Antivirus tools performance in detecting the controlled samples of known steganographic malware embedded in media files from Pexels, a free stock photos repository that allows for use of the photos for such purpose of the research (Pexels.com, 2024). This will be achieved by using Virus Total tool which allows us to determine the detection capabilities without comparing the antivirus from one another (Quintero, 2016). This initial analysis is crucial for establishing a baseline understanding of current detection limitations and identifying areas where improvements are most needed.

Phase 2: Signature Development

Building on identified limitations, this phase will focus on enhancing signature-based detection:

- Creation of custom YARA rules specifically targeting steganographic indicators
- Development of syntactic-based signatures for specific byte patterns associated with steganographic techniques
- Implementation of string and entropy analysis methodologies for file anomaly detection
- Enhancement of static pattern recognition capabilities for embedded code fragments

This rule development will be an iterative process, continuously tested against known samples and newly generated variations to ensure effectiveness and adaptability. This phase focuses on developing specific, pattern-based detection capabilities.

Phase 3: Machine Learning Integration

This phase will combine signature-based approaches with anomaly detection capabilities:

- Development of machine learning models trained to identify statistical anomalies in files
- Training on diverse datasets to ensure robustness across different file types and embedding techniques

- Implementation of detection capabilities for specific anomalies including:
 - Unusual pixel value distributions
 - Entropy pattern variations
 - File structure modifications
 - Metadata inconsistencies

As stated in the concepts of the study, the unsupervised machine learning approach will be evaluated for detecting unknown threats.

Phase 4: Hybrid System Implementation

The final phase will integrate developed components into a detection system:

- Implementation of a two-step detection process combining:
 1. Enhanced YARA-based signature detection for known steganographic techniques
 2. Machine learning-based anomaly detection for identifying previously unknown variants
- Development of system orchestration to optimize workflow and minimize processing overhead
- Implementation of feedback mechanisms to reduce false positives through correlation of signature and anomaly findings
- Optimization of detection thresholds based on empirical testing results

Phase 5: Experimental Setup

The experimental environment will be designed to ensure both security and reproducibility:

- Isolated virtual machines for malware analysis and testing (Ubuntu V24 gnu/Linux and Windows 11)
- Version-controlled implementation to track developmental iterations (GitHub and Git)

This setup will enable safe handling of malicious samples while maintaining experimental integrity and reproducibility.

1.6.2. Data Collection and Sampling

Data collection will involve both existing and newly generated malware samples to ensure comprehensive testing. Sources will include:

1. **Public Malware Repositories:** Samples will be collected from established security research databases such as MalwareBazaar and Contagio Malware Dump.
2. **Generated Steganographic Malware:** Custom samples will be created using various open source stego tools, custom scripts based on LSB principles and AI generation tools to test detection capabilities against threats.
3. **Benign Media Collections:** Clean media files will be gathered to establish baseline characteristics and test for false positives from Pexels.

Security Community Datasets: Collaborative research datasets will be incorporated where available to enhance sample diversity such as Kaggle and various university research projects.

1.6.3. Evaluation Framework

The evaluation framework will assess detection capabilities across multiple sections:

1. **Detection Accuracy:** Measured through true positive rates for known and new steganographic malware samples.
2. **False Positive/Negative Rates:** Assessment of incorrect classifications to gauge practical implementation viability.
3. **Processing Overhead:** Measurement of computational resources required for detection to ensure scalability.
4. **Resilience Against Evasion:** Testing against various complication and anti-detection techniques to evaluate effectiveness.

1.6.4. Tools and Experimental Setup

1.6.4.1. Software Tools

The study will use a combination of established security tools and custom-developed components:

- **YARA Framework:** For rule development, testing, and implementation

- **Virus Total API:** For comparative scanning against multiple antivirus engines
- **Machine Learning Frameworks:** Primarily TensorFlow and PyTorch for model development
- **Steganographic Tools:** For sample generation and embedding techniques the study will make use of Steghide and custom Python scripts

1.6.4.2. Custom Analysis Environment

This is an Isolated testing infrastructure for controlled experimentation which the study will leverage Virtual Box hosted ubuntu Linux/gnu v24 and kali Linux

1.6.5. Expected Outcomes and Contributions

This section outlines the anticipated results and the broader impact of this research. The study is designed to yield academic advancements, enriching the theoretical understanding, and practical contributions, providing tangible benefits to the cybersecurity community. These outcomes collectively aim to address the evolving challenge of AI-generated steganographic malware by offering innovative detection methodologies and accessible solutions.

1.6.5.1. Academic Contributions

The study aims to make significant contributions to the academic understanding of steganographic malware detection. These contributions will expand the theoretical understanding of steganographic threat detection and provide foundations for future research in this evolving field. This includes:

- **Development of a comprehensive framework for analysing AI-generated steganographic malware:** This framework will provide a structured approach for researchers and practitioners to understand, categorize, and analyse the unique characteristics and behaviours of malware hidden using AI-driven steganography. It will cover aspects from malware generation to detection mechanisms, offering a holistic view.
- **Advancement of detection methodologies combining signature-based and anomaly detection approaches:** This involves pushing the boundaries of current detection techniques by demonstrating how YARA rules (signature-based) can be effectively synergized with machine learning anomaly detection to create a more robust and adaptive system capable of identifying both known and novel steganographic threats. This goes beyond simply using them side-by-side by exploring how their strengths can complement each other.

- **Generation of empirical data on detection effectiveness against emerging threat vectors:** The research will produce concrete, measurable data on how well the proposed hybrid system performs against various forms of AI-generated steganographic malware. This empirical evidence will be valuable for validating the new methodologies and informing future research and development in cybersecurity. This data will specifically highlight the system's performance against previously challenging, "emerging" threats.
- **Creation of taxonomies for categorizing and understanding steganographic techniques in malware:** This involves developing a classification system for the different methods and characteristics of steganographic techniques used in malware. Such taxonomies are critical for organizing knowledge, identifying trends, and providing a common language for researchers and analysts to discuss and address these threats more effectively.

These contributions will expand the theoretical understanding of steganographic threat detection and provide foundations for future research in this evolving field.

1.6.5.2. Practical Contributions

Beyond academic advancement, the research will contribute on the practical security enhancements:

- Enhanced YARA rule sets specifically designed for steganographic malware detection
- Machine learning model trained to identify statistical anomalies in files
- Open-source contributions to improve overall cybersecurity protection against emerging threats

These practical outcomes will benefit the broader cybersecurity community by providing applicable methodologies for enhancing detection capabilities.

1.7. Ethical Considerations

It is necessary to ensure ethical integrity during the study to uphold the credibility and dependability of intellectual efforts (Hasan et al., 2021). The ethical issues that underlie this study are discussed in this part, with a focus on permission and informed consent, and confidentiality:

1.7.1. Permission and Informed Consent

The study will not involve any participants, thus no necessary approvals relevant from the institutions and informed consent from participants will be required. The

study will however follow the different software's and applications terms of use, privacy and conditions specifically when it comes to the following applications:

- Virus Total: according to Quintero (2016), Virus Total should not be used to generate comparative metrics between different antivirus products as Antivirus engines can be sophisticated tools that have additional detection features that may not function within the Virus Total scanning environment thus the research will aim to improve the detection capabilities of the antiviruses instead on comparing the detection metrics between them.

Pexels: Pexels license prohibits its content from been sold, redistributed as part of trademark and manipulating people may not appear in a bad light (Pexels.com, 2024) thus this research follows the allowance of modifying of the photos and videos from Pexels without going against the license.

1.8. Proposed Solutions

Given that the traditional antivirus engines struggle to detect most of the malware once it is concealed using steganography (Almomani et al., 2022), this study proposes the development of a hybrid detection model that addresses the gap in the current solutions and the challenge posed by these AI-generated steganography (Gao et al., 2022). The model integrates enhanced signature-based scanning with machine learning-based anomaly detection and custom created YARA rules. The objective is to create a more resilient and adaptive defence against these evolving threats (Chang & Echizen, 2025).

The proposed solution is composed of two primary components:

1.8.1. Enhanced YARA-Based Signature Detection

The first layer of the proposed solution is a specialized set of custom created YARA rules. YARA is a widely used tool for identifying malware based on textual or binary patterns (NCCIC, 2015). However, standard signature-based techniques are often ineffective against polymorphic malware designed to evade detection (Altawil & Youssef, 2023; Mirza et al., 2021). Therefore, this component will focus on developing custom YARA rules that specifically target the subtle artifacts of steganography. This involves creating signatures based on predefined patterns (Rajendran et al., 2019) associated with steganographic tools, including structural file anomalies, unusual entropy levels, and specific byte sequences indicative of hidden data. This component will serve as a fast and efficient first-pass filter to catch known steganographic malware variants.

1.8.2. Machine Learning-Based Anomaly Detection

To overcome the limitations of signature-based detection, the second component is an unsupervised machine learning model designed to detect unknown or zero-day steganographic threats. Unsupervised learning is ideal for this task as it excels at discovering hidden patterns and classifying data without pre-existing labels (Alloghani et al., 2020). This model will function as an anomaly detection system, learning the statistical characteristics of benign image files to identify outliers (Stoian, 2020). It will analyse files for statistical irregularities such as unusual pixel value distributions, entropy pattern variations, and file structure modifications (R.Loganathan, 2024). This allows the system to flag those steganographic techniques that would otherwise go unnoticed.

1.8.3. Integrated Two-Step Pipeline

The final proposed solution is a two-step detection pipeline that combines these components to maximize both efficiency and accuracy of steganography detection.

In short, the model will follow these:

1. **Initial Scan:** All files are first scanned using the custom YARA rule set for a rapid assessment against known indicators.
2. **Deep Analysis:** Files flagged by YARA, or any file requiring deeper inspection, are passed to the machine learning model for in-depth anomaly detection.

This hybrid approach, integrating YARA rules with machine learning, is hypothesized to yield better detection results while reducing false positives (Hammadeh & Kavitha, 2023).

1.9. Chapter Summary.

This section provides a structured overview of the research presented in the preceding chapters, detailing the progression from problem identification to proposed solutions and empirical validation.

1.9.1. Chapter 1: Proposal

Chapter 1 introduces the escalating cybersecurity challenge posed by AI-generated steganographic malware. It establishes the context wherein advanced artificial intelligence, particularly large language models, facilitates the development of steganographic malware that can be concealed within benign digital files, primarily images. The inherent difficulty of detecting such concealed threats with traditional antivirus solutions, which often fail once malware is embedded, is highlighted as a significant problem. The chapter explains the

primary aim of this study which is to evaluate and enhance the detection capabilities of open-source antivirus solutions against AI-generated steganographic malware, with a specific focus on improving YARA's efficacy through the integration of machine learning-based anomaly detection.

1.9.2. Chapter 2: Literature Review

Chapter 2 will systematically review existing literature in steganography and malware detection. It will trace the evolution of steganographic techniques from historical practices to advanced digital methods, emphasizing the shift towards AI-driven concealment. The limitations of current antivirus solutions, particularly their susceptibility to advanced concealment techniques and their reliance on signature-based detection, are thoroughly examined. Furthermore, the chapter will detail the specific challenges associated with detecting steganographic malware, noting the ineffectiveness of existing anti-malware tools against concealed malicious content. The capabilities and inherent limitations of YARA, particularly its struggle against evolving and evasive threats, will also be discussed. Finally, the chapter will explore emerging solutions, such as machine learning-based anomaly detection, as promising sides for enhancing detection capabilities.

1.9.3. Chapter 3: Research Methodology

Chapter 3 will outline the methodological framework employed in this study. It will detail an experimental design utilizing a primarily quantitative approach to measure key performance indicators such as detection rates, false positive ratios, and processing efficiency. The methodology will incorporate a mixed-methods approach, encompassing technical analysis of malware samples, performance evaluation of detection systems, comparative assessment against benchmark datasets, and iterative development of enhanced detection methodologies. Furthermore, the chapter will specify the data collection and sampling strategy, which includes acquiring samples from public malware repositories, generating custom AI-driven steganographic malware, and utilizing benign media collections to establish baseline characteristics.

1.9.4. Chapter 4: Empirical Study

Chapter 4 will describe the practical implementation and experimental phases of the research. It will detail the initial analysis of existing antivirus solutions using tools like Virus Total to establish a baseline of detection capabilities against known steganographic malware. Subsequently, then elaborates on the signature development phase, including the creation of customized YARA rules specifically targeting steganographic indicators, the development of syntactic-based signatures, and the implementation of analysis methodologies. A significant portion

is dedicated to the machine learning integration phase, outlining the development and training of machine learning models to identify statistical anomalies such as unusual pixel value distributions, entropy variations, and file structure modifications. The chapter will conclude by describing the hybrid system implementation, integrating the enhanced YARA-based detection with the machine learning anomaly detection into a two-step process designed to maximize detection and minimize false alarms. The software tools and experimental setup, including the use of YARA Framework, VirusTotal API, TensorFlow, PyTorch, and isolated virtual machines, will also be detailed.

1.9.5. Chapter 5: Research Findings

Chapter 5 will present the empirical findings derived from the experimental study. Further detailing the performance of the developed hybrid detection system against AI-generated steganographic malware samples. It will quantitatively present the achieved detection rates, false positive ratios, and processing efficiency, providing a comprehensive assessment of the system's effectiveness. The chapter will also offer comparative analyses, benchmarking the performance of the proposed hybrid model against existing solutions and conventional detection methods.

1.9.6. Chapter 6: Conclusion

Chapter 6 will detail the core contributions and implications of the research. It will show how the study successfully addresses the identified gap in current antivirus solutions' ability to detect AI-generated steganographic malware. The chapter will further summarize the effectiveness of the developed hybrid model, which combines customized YARA rules with machine learning-based anomaly detection, in enhancing detection rates and reducing false positives against sophisticated steganographic techniques. Emphasising in the dual impact of the research, contributing to both theoretical understanding through the development of a comprehensive analysis framework and practical through enhanced open-source detection methodologies and finally discuss the significance of these findings for the cybersecurity community and outlines potential directions for future research in this continuously evolving threat landscape

1.10. Summary

The proposed research addresses a gap in current AV solutions in detecting Ai-generated steganography malware by developing enhanced methodologies for detecting these malwares. Through a hybrid approach combining custom YARA rules and machine learning-based anomaly detection, this study aims to improve detection rates against current stegomalware techniques that currently evade conventional antivirus solutions.

The study will maintain a focus on academic advancement and practical application, ensuring that findings contribute to both theoretical understanding and real-world improvements. By targeting open-source solutions, the research outcomes will be accessible to the broader cybersecurity community, maximizing potential impact and adoption.

This research represents a timely and necessary response to this evolving challenge, with potential benefits for organizations across sectors and individual users alike.

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